Counting

Basic Counting principles

Sum Rule: if a Task can be done in n_1 ways and a second task in n_2 ways, and if these two tasks can not be done at the same time, then there are $n_1 + n_2$ ways to do either task

Ex: The department will award a free computer to either a CS student or a CS professor.
How many different choices are there if there are 530 students and 15 professors

Ans: $530 + 15 = 545$ choices

product rule: suppose that a procedure can be broken down into two successive tasks. If there are n_1 ways to do the first task and n_2 ways to do the second task after the first task has been done, then there are $n_1 \cdot n_2$ ways to do the procedure

Ex: how many different licence plates are there that contain exactly three English letters

Ans:- $26 \times 26 \times 26 = 17576$

permutation: It's an ordered arrangements of objects

the No. of permutations of r distinct objects chosen from n distinct objects is denoted by $P(n,r)$

$$P(n,r) = \frac{\lfloor n}{\lfloor n-r}$$

permutations of objects is a sequence that contains each object exactly once.

Ex: suppose you have time to listen to 10 songs on your daily jog around campus. There are 6 cake tunes, 8 moby tunes, and 3 eagle tunes to choose from

$$P(17,10) = 17 \times 16 \times 15 \times 14 \times 13 \times 12 \times 11 \times 10 \times 9 \times 8$$

suppose you want to list 4 cake, 4 moby, and 2 eagles in the order

$$P(6,4) \times P(8,4) \times P(3,2)$$

if order does not matter then.

$$P(6,4) \times P(8,4) \times P(3,2) \times \lfloor 3$$

Combination: It's an unordered selection of elements from the same set

the No. of combinations of r distinct objects chosen from n distinct objects is denoted by ${}^{n}C_{r}$

$${}^{n}C_{r} = P(n,r)/\lfloor r = \frac{\lfloor n}{\lfloor r \lfloor n-r}$$

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A basketball squad consists of 12 players, 5 of which make a team. How many different teams of players can you make from 12

$\underline{^{12}C_5}$

WHAT IS THE DIFFERENCE

In a running race of 12 sprinters, each of the top 5 finishers receives a different medal. How many ~~diff~~ ways are there to award the 5 medals.

$^{12}P_5$

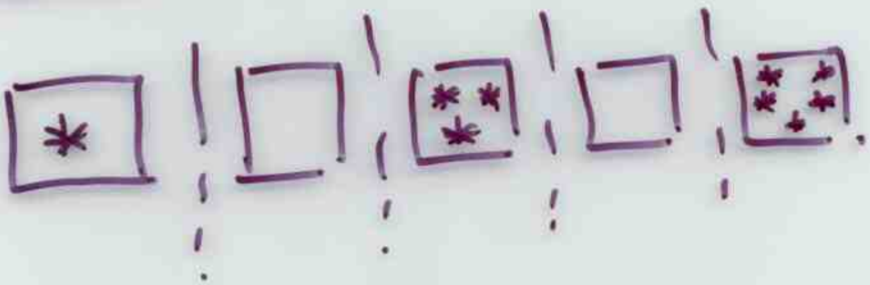
$= {}^{12}C_5 \times \lfloor 5$

permutation with repetitions:-

Thm:- The no. of r permutation of an n element set S with repetition is n^r.

Combination with repetition:-

Consider a college having 5 courses. Find the no. of ways of selecting 9 students from 5 courses when there is no restriction on the number of students to be chosen from each course.



1 0 3 0 5

Now any 9 combination with repetition can be represented as a 13-bit binary sequence. Such that 1 represent the partition and 0 represent student

This sequence has nine 0's and four 1's.

0 1 0 1 0 1 0 0 0 0 1 0 0.

1 1 1 4 2 = 9 students

Thm:- The No. of r combinations with repetition from a set S of n elements is $^{n+r-1}C_r$.

OR.
The No. of distributing r similiar objects in n boxes = the number of binary sequences having $(n-1)$ 1's and r 0's.

Proof: length of sequence:
$(n-1+r)$

Note: 1 and 0 can appear anywhere.

$\therefore$ $^{n+r-1}C_r$.

Pascal's Triangle

In Pascal's triangle, each number is the sum of the numbers to its upper left and upper right

$$
\begin{array}{ccccccccc}
 & & & & 1 & & & & \\
 & & & 1 & & 1 & & & \\
 & & 1 & & 2 & & 1 & & \\
 & 1 & & 3 & & 3 & & 1 & \\
1 & & 4 & & 6 & & 4 & & 1
\end{array}
$$

Since we have $C(n+1, k) = C(n, k-1) + C(n, k)$ and $C(0,0) = 1$ we can use Pascal's triangle to simplify the computation $C(n, k)$.

$$C(0,0) = 1$$

$$C(1,0) = 1 \quad C(1,1) = 1.$$

$$C(2,0) = 1 \quad C(2,1) = 2 \quad C(2,2) = 1$$

$$C(3,0) = 1 \quad C(3,1) = 3 \quad C(3,2) = 3 \quad C(3,3) = 1$$

$$C(4,0) = 1 \quad C(4,1) = 4 \quad C(4,2) = 6 \quad C(4,3) = 4 \quad C(4,4) = 1$$

Binomial Coefficients

A binomial expression is the sum of two terms, such as $(a+b)$

Consider $(a+b)^2 = (a+b)(a+b)$

When expanding such expressions, we have to form all possible products of a term in the first factor and a term in second factor $(a+b)^2 = a \cdot a + a \cdot b + b \cdot a + b \cdot b$

$(a+b)^3 = (a+b)(a+b)(a+b)$
$= aaa + aab + aba + abb + baa + bab + bba$
$+ bbb.$

There is only one term a^3, because there is only one possibility to form it
Choose a from all the three factors.

$C(3,3)=1$

There is ~~three~~ times the term a^2b, because there are three possibilities to choose a. subset of two out of three factors.

$C(3,2)=3.$

~~Simp~~ similarly, there is three times the term ab^2 $C(3,1)=3$ and once the term b^3 $(C(3,0)=1)$

$$(a+b)^n = \sum_{j=0}^{n} C(n,j) \cdot a^{n-j} \cdot b^j$$

[Binomial theorem]

Fifth row of Pascal's triangle
$1-4-6-4-1$ helps us to compute

$(a+b)^4 = a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4.$

sum and the product rules can also be phrased in terms of set theory.

Sum Rule: let $A_1, A_2, \ldots A_m$ be disjoint sets. then the No. of ways to choose any element from one of these sets is
$|A_1 \cup A_2 \cup \cdots \cup A_m| = |A_1| + |A_2| + \cdots + |A_m|$

product rule: let $A_1, A_2 \ldots, A_m$ be finite sets. Then the number of ways to choose one element from each set in the order $A_1, A_2 \ldots, A_m$ is
$|A_1 \times A_2 \times \cdots \times A_m| = |A_1| \cdot |A_2| \cdots |A_m|$.

Inclusion - Exclusion.

How many bit strings of length 8 either start with a 1 or end with 00?

Task 1: - construct a string of length 8 That starts with 1.
product rule: $1 \times 2^7 = 128$ ways.

Task 2: Construct a string of length 8 That ends with 00
product rule: $2^6 = 64$ ways

How many strings start with 1 and end with 00

$$2^5 = 32$$

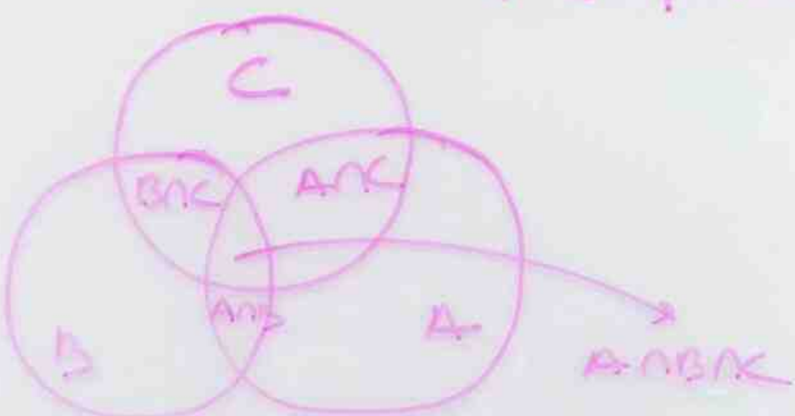
$$\underbrace{128}_{\text{Task 1}} + \underbrace{64}_{\text{Task 2}} - \underbrace{32}_{\substack{\text{Task 1 and} \\ \text{Task 2 at same time}}} = 160 \text{ ways}$$

In set theory, this corresponds to sets A_1 and A_2 that are not disjoint. Then

$$\underbrace{|A_1 \cup A_2| = |A_1| + |A_2| - |A_1 \cap A_2|}_{\text{principle of inclusion - Exclusion}}$$

Thm #2

$$P(A_1 \cup A_2 \cup A_3)$$
$$= P(A_1) + P(A_2) + P(A_3) - P(A_1 \cap A_2) - P(A_1 \cap A_3) - P(A_2 \cap A_3) + P(A_1 \cap A_2 \cap A_3)$$



Thm #3

$$\left| \bigcup_{i=1}^{n} A_i \right| = \sum_{i=1}^{n} |A_i| - \sum_{i,j:1\le i\le j\le n} |A_i \cap A_j| + \sum_{i,j,k:1\le i<j<k\le n} |A_i \cap A_j \cap A_k| - \cdots + (-1)^n |A_1 \cap \cdots A_n|$$

Principle of inclusion Exclusion
(PIE)

* → let S be a finite set and let A, B, C be subsets of S Then

$$|A \cup B| = |A| + |B| - |A \cap B|$$

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

in general

$$|A_1 \cup A_2 \cup \cdots \cup A_n] = \sum_{i} |A_i| - \sum_{i<j} |A_i \cap A_j| + \sum_{i<j<k} |A_i \cap A_j \cap A_k| - \cdots + (-1)^{n+1} |A_1 \cap A_2 \cap \cdots \cap A_n|.$$

Proof :-

$$\underbrace{|A_1 \cup A_2 \ldots \cup A_n|}_{\text{each element is counted extactly once.}} = \sum_{1 \le r \le n} |A_r| - \sum_{1 \le r < s \le n} |A_r \cap A_s|$$

$$+ \sum_{1 \le r < s < t \le n} |A_r \cap A_s \cap A_t| - \ldots + (-1)^{n+1} |A_1 \cap A_2 \cdots \cap A_n|$$

Consider R.H.S

Let $x \in A_1 \cup A_2 \ldots A_n$ Then one of the following n cases occur

Case 1: x lies in exactly one set among A_i's

Case 2: x lies in exactly two sets among A_i's

$\vdots$

Case m: x lies in exactly m sets among A_i's

$\vdots$

Case n: x lies in $A_1 \cap A_2 \ldots \cap A_n$.

Consider case m :-

— It's counted m times in

$\sum |A_r|$

— x occurs in second term of RHS in ${}^mC_2$ ways. (so excluded ${}^mC_2$ times)

— excluded ${}^mC_3$ times

$\Rightarrow m - {}^mC_2 + {}^mC_3 + \cdots + (-1)^{n-1}\, {}^mC_n.$

$\Rightarrow {}^mC_0 - [{}^mC_0 - {}^mC_1 + {}^mC_2 - \cdots + (-1)^m\, {}^mC_n]$

$\Rightarrow 1 - (1-1)^m$ [by binomial]

$\Rightarrow 1$

$\therefore$ LHS = R.H.S

Find the no. of integral solutions of eqn

$$x_1 + x_2 + x_3 + x_4 = 15$$

$$2 \leq x_1 \leq 6 \quad -2 \leq x_2 \leq 1$$

$$0 \leq x_3 \leq 6 \quad 3 \leq x_4 \leq 8$$

Let $y_1 = x_1 - 2 \quad y_3 = x_3 \quad y_2 = x_2 + 2$

$y_4 = x_4 - 3$

$$\Rightarrow \quad y_1 + y_2 + y_3 + y_4 = 12$$

subject to $0 \leq y_1 \leq 4 \quad 0 \leq y_2 \leq 3$

$0 \leq y_3 \leq 6 \quad 0 \leq y_4 \leq 5$

Let S be the ~~set~~ of all nonnegative integral solutions of the eqn.

$$y_1 + y_2 + y_3 + y_4 = 12$$

Let $P_1: \; y_1 \geq 5 \quad P_2: y_2 \geq 4 \quad P_3: y_3 \geq 7$

$P_4: y_4 \geq 6$

Soln: Cardinality of $|\overline{A_1} \cap \overline{A_2} \cap \overline{A_3} \cap \overline{A_4}|$

$|S| = {}^{12+4-1}C_{12} = {}^{15}C_{12} = 455$

$|A_1| = {}^{7+4-1}C_7 = {}^{10}C_7 = 120$

$|A_2| = {}^{8+4-1}C_8 = {}^{11}C_8 = 165$

$|A_3| = {}^{5+4-1}C_5 = {}^{8}C_5 = 56$

$|A_4| = {}^{4+6-1}C_6 = {}^{9}C_6 = 84$

<u>Now</u>

$|A_1 \cap A_2| = {}^{4+3-1}C_3 = {}^{6}C_3 = 20$

$|A_1 \cap A_3| = 1 \quad |A_1 \cap A_4| = 4$

$|A_2 \cap A_3| = 4 \quad |A_2 \cap A_4| = 10$

$|A_3 \cap A_4| = 0$

<u>for the intersection of more sets</u>

$|A_1 \cap A_2 \cap A_3| = |A_1 \cap A_2 \cap A_4|$

$= |A_1 \cap A_3 \cap A_4|$

$= |A_2 \cap A_3 \cap A_4|$

$= |A_1 \cap A_2 \cap A_3 \cap A_4| = 0$

$\therefore |\overline{A_1} \cap \overline{A_2} \cap \overline{A_3} \cap \overline{A_4}|$

$\equiv 455 - (120 + 165 + 56 + 84) + (20 + 1 + 4 + 4 + 10)$

$= 69$

15

Hat check Problem —:

$n = 5$ people check in their hats. Later they are given back their hats. Find the No. of ways that no person receives his/her hat

5 objects: 1, 2 3 4 5

U: # of permutations = $\lfloor 5$

A: person 1 recieves his hat # permutation $= \lfloor 4$

" 2 " " " ... .

rows: $5 = C(5,1) \cdot |A| = C(5,1)\lfloor 4$

B: person 1 & 2 recieve their hats # permutation $= \lfloor 3$

person 1 & 3 " ... .

rows: $(5,2) \cdot |B| = C(5,2)\lfloor 3$

C: person 1, 2 & 3 recieve their hats # permutation $= \lfloor 2$

person 1, 2 & 4 recieve their hats # permutation ....

rows $C(5,3) \cdot |C| = C(5,3)\lfloor 2$

D: person 1, 2, 3 & 4 recieve their hats # permutation $= \lfloor 1$.

person 1, 2, 3 & 5 - " ... .

rows $= C(5,4) \cdot |D| = C(5,4)\lfloor 1$

E: Person 1, 2, 3, 4 & 5 recieve their hats # permutation $= 1$.

Total # permutations

$$- n(A) + n(B) - n(C) + n(D) - n(E)$$

$$= \underline{|5} - \underline{|5} + \frac{\underline{|5}}{\underline{|2}} - \frac{\underline{|5}}{3} + \frac{\underline{|5}}{4} - \frac{\underline{|5}}{\underline{|5}}$$

$$= \underline{|5}\left[1 - 1 + \frac{1}{\underline{|2}} - \frac{1}{\underline{|3}} + \frac{1}{\underline{|4}} - \frac{1}{\underline{|5}} \right]$$

$=$

include E I E I E